

Here are 10 Code Challenges for ES6 Features

1. Replace `var` with `let` and `const`

- Rewrite the following code using `let` and `const` instead of `var`:

```
var name = "Alice";  
var age = 25;  
var isStudent = true;
```

2. Arrow Function to Calculate Sum

- Write an arrow function that takes two numbers as parameters and returns their sum.

3. Use Arrow Function for String Concatenation

- Write an arrow function that accepts two strings and returns their concatenation.

4. Convert Callback to Arrow Function

- Given the following callback function, convert it to an arrow function:

```
function greet(name) {  
  console.log("Hello, " + name);  
}
```

5. Use `let` and `const` in Loops

- In the following code, replace `var` with `let` or `const` where appropriate:

```
var i;  
for (i = 0; i < 5; i++) {  
  console.log(i);  
}
```

6. Create a Simple Class with Constructor

- Create a class `Person` with a constructor that takes `name` and `age` as parameters and prints them to the console when a new `Person` object is created.

7. Inheritance in JavaScript

- Create a `Vehicle` class with properties `make` and `model`, and a method `drive()`. Then, create a `Car` subclass that inherits from `Vehicle` and adds a method `honk()` to print "Honk!".

8. Destructuring Assignment for Arrays

- Given the following array, use **array destructuring** to extract the first and second values into separate variables:

```
const numbers = [10, 20, 30];
```

9. Destructuring Assignment for Objects

- Given the following object, use **object destructuring** to extract the `name` and `age` properties into separate variables:

```
const person = { name: "John", age: 30, city: "New York" };
```

10. Template Literals for Dynamic Strings

- Write a function that takes a `name` and an `age` as arguments and returns a string using **template literals**: "My name is [name] and I am [age] years old.".